

Sparse Minimality of Gaussian-Moments Counterexamples in Dimension Three

Roy van Rijn
Independent researcher

24 July 2026

Abstract

The Gaussian Moments Conjecture fails in every dimension at least three. We study minimal failure in the first failing dimension. An exhaustive exact search over all 59,535 supports having degree at most four and at most four monomials shows that every pure-moment-null polynomial in this range still satisfies the conjectural mixed-moment conclusion. Thus Long's five-term quartic is Pareto-minimal with respect to degree and circular-coordinate support size. The theorem is computer-assisted: moment ideals through order twenty-four are computed directly in Laurent coordinate rings over $\mathbb{Q}$, and the three genuine surviving families are proved harmless at all orders.

1 Introduction

Let $X_1, \dots, X_d$ be independent standard real Gaussian random variables. The Gaussian Moments Conjecture, introduced by Derksen, van den Essen, and Zhao [1], asserts that if

$$P \in \mathbb{C}[X_1, \dots, X_d], \quad \mathbb{E}(P^m) = 0 \quad (m \geq 1),$$

then, for every $Q \in \mathbb{C}[X_1, \dots, X_d]$,

$$\mathbb{E}(QP^m) = 0 \quad \text{for all sufficiently large } m.$$

Long recently found explicit counterexamples in every dimension $d \geq 3$ [2]. His three-variable witness, in circular coordinates W, Z and an independent real Gaussian T , is

$$P_L = W + WZ - T^2 - \frac{3}{2}ZT^2 - \frac{1}{2}Z^2T^2. \tag{1}$$

It has degree four and five monomials, all its pure moments vanish, and $\mathbb{E}(ZP_L^m) = m!$ for every $m \geq 1$. We place this witness at the first corner of the sparse degree–support plane.

Theorem 1.1 (Sparse Pareto minimality). *Let W, Z, T be Gaussian Witt coordinates, so that*

$$\mathbb{E}(W^i Z^j T^k) = \begin{cases} i!(k-1)!!, & i = j \text{ and } k \text{ is even,} \\ 0, & \text{otherwise.} \end{cases}$$

Suppose that $P \in \mathbb{C}[W, Z, T]$ has total degree at most four and at most four nonzero monomials. If $\mathbb{E}(P^m) = 0$ for every $m \geq 1$, then, for every $Q \in \mathbb{C}[W, Z, T]$,

$$\mathbb{E}(QP^m) = 0 \quad \text{for all sufficiently large } m.$$

Consequently no GMC counterexample has both degree at most four and support size at most four in a Gaussian Witt frame.

In particular, (1) is minimal for the product partial order on $(\deg P, \# \text{Supp}(P))$. The theorem does not separately prove that degree four is globally minimal: a denser cubic is not excluded. Nor does it exclude a four-term counterexample of degree greater than four.

2 Moment ideals

The finite part of the proof is an exhaustive exact coefficient-scheme calculation; the all-order interpretation of its survivors is conceptual. Let

$$\mathcal{M}_4 = \{W^i Z^j T^k : i, j, k \geq 0, i + j + k \leq 4\}.$$

This set has 35 elements. Fix a support $S \subset \mathcal{M}_4$, write

$$P_S = \sum_{s \in S} c_s s,$$

and work on the coefficient torus $\prod_{s \in S} c_s \neq 0$. For $s = (i_s, j_s, k_s)$ and a multiplicity vector $\nu = (\nu_s)_{s \in S} \in \mathbb{N}^S$, put

$$|\nu| = \sum_s \nu_s, \quad I(\nu) = \sum_s \nu_s i_s, \quad J(\nu) = \sum_s \nu_s j_s, \quad K(\nu) = \sum_s \nu_s k_s.$$

Before expanding any Wick coefficients, associate to S its contraction semigroup

$$\Gamma_S = \left\{ \nu \in \mathbb{N}^S : \sum_s \nu_s (i_s - j_s) = 0, \quad \sum_s \nu_s k_s \equiv 0 \pmod{2} \right\}.$$

Its Hilbert basis H_S consists of the primitive contractions. This basis is computed exactly in two stages. First ignore parity. Pottier's bound [3] shows that every Hilbert-basis vector for the one-row charge equation has

$$|\nu| \leq 1 + \sum_{s \in S} |i_s - j_s|.$$

The even-parity subsemigroup is then generated by the even charge atoms and the pairwise sums, with repetition, of odd charge atoms; removing decomposable candidates gives H_S .

Wick contraction gives the exact moment polynomial

$$\mathbb{E}(P_S^m) = \sum_{\substack{|\nu|=m, \\ I(\nu)=J(\nu) \\ K(\nu) \text{ even}}} \frac{m!}{\prod_s \nu_s!} I(\nu)! (K(\nu) - 1)!! \prod_s c_s^{\nu_s}. \quad (2)$$

Thus the truncated full-support coefficient scheme is most naturally defined directly on the coefficient torus by the Laurent ideal

$$J_{S,M} = \langle \mathbb{E}(P_S), \dots, \mathbb{E}(P_S^M) \rangle \subseteq \mathbb{Q}[c_s^{\pm 1} : s \in S]. \quad (3)$$

The equations are homogeneous, so one coefficient may be normalized to one. There is a second coefficient symmetry. The Gaussian-preserving torus

$$W \mapsto tW, \quad Z \mapsto t^{-1}Z, \quad T \mapsto T$$

acts on the coefficient of $s = W^{i_s} Z^{j_s} T^{k_s}$ with charge $q_s = i_s - j_s$. Thus the coefficient torus carries the action

$$(\lambda, t) \cdot c_s = \lambda t^{q_s} c_s,$$

and every moment equation is invariant under t and homogeneous under λ .

If a support contains coefficients c_a, c_b of distinct charges, then over an algebraic closure every orbit meets the slice $c_a = c_b = 1$: one solves

$$t^{q_b - q_a} = c_a / c_b, \quad \lambda = (t^{q_a} c_a)^{-1}.$$

Equivalently, with $d = q_b - q_a$, quotient coordinates are the invariant Laurent ratios

$$y_s = c_s^d c_a^{q_s - q_b} c_b^{q_a - q_s} \quad (s \neq a, b).$$

On the slice they become $y_s = c_s^d$, so the slice is a finite cover of the invariant-ratio quotient. Consequently the four-term elimination has only two active coefficient variables whenever two charges occur. If all charges agree, the torus acts through overall scaling; such an active non-one-sided support necessarily has charge zero, and the calculation falls back to one coefficient normalization.

There are

$$\sum_{r=1}^4 \binom{35}{r} = 59,535$$

raw supports. We identify a support with its reflection under $W \leftrightarrow Z$. Two elementary filters precede the Groebner calculation. If H_S has one element, then Γ_S has rank one and the moment at the degree of its generator consists of a single nonzero coefficient monomial; the coefficient torus contains no solution. More generally, the contraction vectors are enumerated by total degree from H_S , and a degree containing exactly one vector gives the same rejection before any factorial Wick weights are expanded. Supports with the same ordered charge/parity signature share this enumeration. Only the remaining supports are expanded using (2). If no contraction occurs, all charges $i_s - j_s$ have one strict sign and the mixed-moment conclusion follows by bounded weight. Possible additional factorizations among disjoint primitive contractions are not used in the certificate.

For completeness, the latter test is finite in this range. A zero charge contracts in one or two copies, according to its T -parity. Opposite nonzero charges $q > 0 > r$, with $|q|, |r| \leq 4$, have a primitive zero-sum relation of length

$$\frac{q + |r|}{\gcd(q, |r|)} \leq 8;$$

doubling corrects odd T -parity. Hence every support which is not strictly one-sided has a contraction by order sixteen.

3 Exact enumeration

After the preceding filters, the four-term stratum contains 5,718 coefficient-torus ideals requiring elimination. Each is computed in the Laurent ring of the two-normalization slice when its charge data have rank two, with the one-normalization fallback just described. No inverse or Rabinowitsch variable is adjoined: extension to the Laurent ring and contraction are performed by ideal quotients in the genuine quotient coordinates. Exact Groebner bases over $\mathbb{Q}$ for $J_{S,20}$ leave precisely the supports

$$\begin{aligned} &\{Z, ZT^2, WZ^2, W^3T\}, \\ &\{ZT, Z^2, WT, W^2\}, \\ &\{ZT, ZT^3, WZ^2T, W^3\}. \end{aligned}$$

The first and third acquire the unit ideal after adjoining moments through order twenty-four. The same exact search in the one-, two-, and three-term strata leaves no active one- or two-term family and two three-term families. Including the four-term survivor, the genuine active null families are

$$P_{\text{lin}} = T + aZ + bW, \quad 1 + 2ab = 0, \quad (4)$$

$$P_{\text{sh}} = Z + aWT - \frac{1}{2}a^2W^3, \quad (5)$$

$$P_{\text{pr}} = ZT + aZ^2 + \frac{1}{2a^2}WT - \frac{1}{4a^3}W^2, \quad a \neq 0. \quad (6)$$

For the three survivors only, clearing Laurent monomials exports triangular lexicographic certificates. With the first coefficient normalized to one, their nontrivial equations are respectively

$$1 + 2c_1c_2, \quad 2c_2 + c_1^2, \quad 2c_1^2c_2 - 1, \quad c_3 + c_1c_2^2.$$

These formulas both classify the coefficient schemes and make the survivors easy to interpret.

4 All-order interpretation of the survivors

The family (4) is an isotropic linear form. A complex orthogonal change takes it to a scalar multiple of Z . A fixed polynomial Q has bounded W -degree, so $\mathbb{E}(QZ^m) = 0$ for all sufficiently large m .

For (5), define

$$D = W\partial_T - T\partial_Z.$$

Then $D(2WZ + T^2) = 0$, $D(W) = 0$, and

$$P_{\text{sh}} = \exp(-aWD)(Z).$$

Gaussian integration by parts gives $\mathbb{E}(DF) = 0$ for every polynomial F , and therefore also $\mathbb{E}(WDF) = 0$. Since WD is locally nilpotent, $\phi = \exp(-aWD)$ is a polynomial automorphism preserving the Gaussian functional. For fixed Q ,

$$\mathbb{E}(QP_{\text{sh}}^m) = \mathbb{E}(\phi^{-1}(Q)Z^m),$$

which vanishes eventually by the same bounded-weight argument.

Finally,

$$P_{\text{pr}} = \left(Z + \frac{1}{2a^2}W\right) \left(T + aZ - \frac{1}{2a}W\right). \quad (7)$$

The second factor is isotropic and orthogonal to the first. After an orthogonal Witt change, (7) is a scalar multiple of TZ . Every P_{pr}^m then has isotropic weight m , which a fixed Q can balance only finitely often. All three families satisfy GMC. Together with the strictly one-sided supports, this proves Theorem 1.1.

5 Reproducibility and scope

The enumeration is implemented with exact integer Wick coefficients and Singular’s rational Groebner bases. It computes each contraction Hilbert basis before moment expansion, caches contraction vectors by charge/parity signature, and enumerates supports rather than reading a stored survivor list. It performs every emptiness test over $\mathbb{Q}$ in the Laurent ring of the coefficient-symmetry quotient, without auxiliary saturation variables. Polynomial denominators are cleared only to export the three displayed triangular certificates. The replay also verifies the invariant-ratio exponents and checks (7). It is run from the repository root by

```
.venv/bin/python scripts/verify_three_real_gmc_sparse_pareto.py
```

The complete replay takes under a minute on a laptop. Modular runs are used only as optional discovery filters and are not part of the exact certificate.

Acknowledgments and declaration of generative AI use

The author used OpenAI’s ChatGPT and Codex extensively during the research process, including for conjecture formation, symbolic experimentation, proof exploration, consistency checks, and preparation of this manuscript. The author reviewed the mathematical claims and assumes full responsibility for the contents. The theorem uses the exhaustive exact coefficient-scheme calculation stated in the text and supplied with the paper.

References

- [1] Harm Derksen, Arno van den Essen, and Wenhua Zhao. The Gaussian Moments Conjecture and the Jacobian Conjecture. *Israel Journal of Mathematics*, 219(2):917–928, 2017. DOI.
- [2] Christopher D. Long. Small counterexamples to the Gaussian Moments Conjecture, 2026. arXiv:2607.18186.
- [3] Loïc Pottier. Minimal solutions of linear diophantine systems: Bounds and algorithms. In *Rewriting Techniques and Applications*, volume 488 of *Lecture Notes in Computer Science*, pages 162–173. Springer, 1991. DOI.